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Empowering Structural Integrity: YOLO-Based Crack Detection and ML- Driven Concrete Strength Prediction of Critical Infrastructure Caused due to Mining Operation.



Abstract: - In this paper, YOLO algorithm is used for detecting crack by incorporating machine learning approach utilized for predicting concrete strength and possible crack in critical infrastructure caused during mining operation. Over time, those fissures become more noticeable until they finally collapse. Big data is accumulating huge amount of data by utilizing remote sensing, earth penetrating RADAR, LiDAR, GPS, and satellite image data for large-scale operations. The performance of the proposed approach is compared with other state-of-the-art algorithms, such as, YOLO V3 and YOLO V8. To investigate the performance, four performances metric like mean absolute error (MAE), root mean square error (RMSE), relative standard error (RSE) and root relative squared error (RRSE) has considered. It is found that our approach is giving superior results in term of the compared performance metrics. Moreover, non-parametric statistical testing, such as, signed test, Wilcoxon Rank test and Friedman's test has been conducted to verify the dominance of proposed approach over others.

Keywords: fissure, RADAR, RMSE, YOLO V8, LiDAR

1. Introduction:

The detection of cracks in critical infrastructure, especially in structures like bridges, is of paramount importance due to the potential for catastrophic consequences resulting from undetected structural defects. Cracks serve as indicators of structural weaknesses that, if neglected, can compromise the entire infrastructure's integrity. Our emphasis on evaluating these models, with Yolov8 showcasing exceptional performance, underscores the critical need for leveraging advanced technologies to accurately and promptly identify structural vulnerabilities.

In the domain of structural monitoring, the integration of Internet of Things (IoT) technologies assumes paramount importance. While visible inspections and manual measurements are crucial, they may not capture the nuanced complexities of structural deformities. The implementation of IoT devices facilitates continuous, real-time monitoring of infrastructure, providing a comprehensive and dynamic understanding of its condition. Utilizing various methods, ranging from traditional visual inspections to advanced technologies like Ground Penetrating Radar (GPR) and Acoustic Emission Monitoring, ensures a multi-faceted approach to identifying potential risks. This integration not only enhances the accuracy of crack detection but also enables proactive and targeted maintenance, mitigating the potential for disasters.

While crack detection is indispensable, predicting the strength of concrete adds an additional layer of foresight crucial for preventative measures. Estimating concrete strength aids in identifying structural vulnerabilities before they manifest as visible cracks. The urgency of predictive capabilities is underscored by incidents resulting from structural failures due to undetected defects. Machine learning models, including ZeroR, RBFR, Kstar, and RF. These predictive models not only enhance our understanding of structural health but also provide a basis for strategic decision-making, reducing the likelihood of incidents resulting from structural weaknesses.

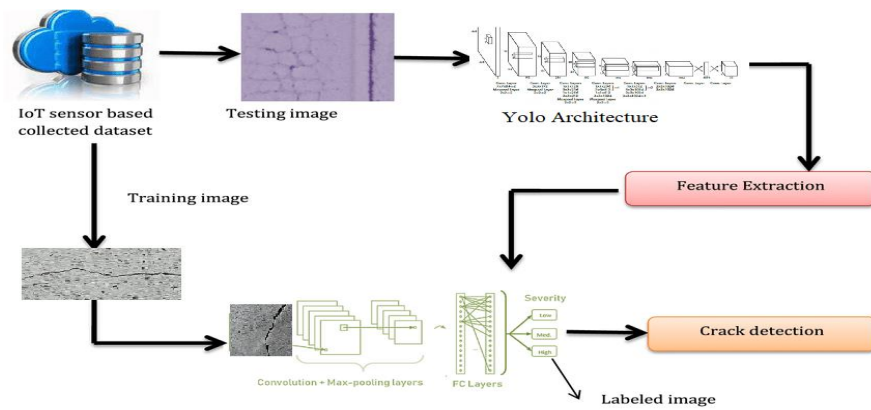


Fig.1. Flow of crack detection

Our research signifies a critical advancement in infrastructure monitoring and maintenance. The integration of YOLO models for crack detection, along with IoT-based monitoring and predictive modeling of concrete strength, addresses the pressing need for a comprehensive and proactive approach to structural integrity. This integrated approach holds immediate implications for the safety and longevity of critical infrastructure and sets a precedent for the continued evolution of technology-driven solutions in civil engineering. As we strive to build safer and more resilient structures, our research stands as a pioneering effort that bridges the gap between traditional inspection methods and cutting-edge technologies, ensuring the structural well-being of critical infrastructure.

2. Related Work

Dung C. V. et al. [1] presents a novel approach employing a fully convolutional network (FCN) for substantial break detection. It leverages three unique pre-prepared network models for image classification. Through training on a subset of clarified pictures, the method attains an impressive 90% average precision in semantic segmentation. Validation on a cyclic loading test video illustrates its proficiency in detecting cracks and accurately assessing crack density, underscoring its robustness in civil engineering structure monitoring. Fang F. et al [2] introduces an innovative methodology for break identification in crude pictures, addressing the challenges posed by weak signals in noisy backgrounds. The methodology consolidates profound learning models with Bayesian probabilistic investigation to enhance robustness. A state-of-the-art object finder is re-trained to recognize break patches with appropriate signal-noise-ratio (SNR), and a self-loader strategy generates ground bits of insight for training. A Bayesian integration algorithm further refines crack detections, suppressing false positives by considering spatial nearness, direction consistency, and arrangement consistency. Extensive experiments on a new dataset showcase the method's superiority over existing approaches, with ablation experiments confirming the adequacy of the proposed strategies.

Zhang L. et al. [3] proposes CrackUnet, a deep learning model, to overcome these challenges. A diverse dataset is manually labeled, augmented, and resized to $256 \times 256 \times 3$ dimensions, yielding 1200 sub images. The CrackUnet model, an enhanced Unet-based method, employs a generalized dice loss function for precise pixel-level crack detection. Evaluation on test and Crack Forest datasets showcases superior performance, achieving up to 91.45%, 88.67%, and 90.04% precision, recall, and F1 score, respectively. Comparative analyses underscore CrackUnet's superiority in terms of accuracy, training time, and dataset characteristics. Robustness and generalization are validated through testing on noisy images, affirming its efficacy in concrete structure assessment. Xu H. et al. [4] presents an innovative start to finish break discovery model utilizing a convolutional neural network (CNN). Exploiting atrous convolution, Atrous Spatial Pyramid Pooling (ASPP) module, and profundity wise distinguishable convolution, the model attains a notable 96.37% detection accuracy without pre-training. Outperforming traditional classification models, it offers adaptable integration into diverse convolutional networks for effective feature extraction, showcasing its potential for automated and precise concrete bridge crack detection, crucial for ensuring transportation safety.

Ali R. et al. [5] review examines CNN execution in common design break identification, highlighting advancements in equipment, information assortment, and calculations. Focusing on picture pre-handling, equipment, programming apparatuses, datasets, network models, learning methods, misfortune capabilities, and execution metrics, the review critically analyzes recent developments. It also discusses manual processes, picture handling procedures, and AI techniques, acknowledging their restrictions. The article points to provide per users with insights into CNN-based break discovery techniques and stimulate exploration of future research challenges. Dinh T. H. et al. [6] addresses challenges in vision-based substantial break discovery through a unique integration of break mechanics, recreation, and AI. Utilizing machine learning on simulation results, the approach develops mathematical models for concrete cracks, thereby improving accuracy in detection models. Experiments conducted on different break picture datasets reveal a strong connection among recreated and genuine-world cracks. The innovative methodology significantly enhances accuracy in cutting edge identification and division models, demonstrating a notable 1.27% average improvement across taking part models and a remarkable 5.47% boost on U-Net. This interdisciplinary strategy addresses crucial concerns in a research domain where data quality and quantity are pivotal for advancements.

Aaron Rasheed Rababaah [7] explores the efficacy of a Convolutional Neural Network (CNN) in black-top asphalt break location, departing from traditional Machine Learning (ML) methods. The study introduces a deep learning (DL) model, demonstrating its effectiveness through a dataset of 500 images with a notable mean precision of 96% and a standard deviation of 0.025. The paper contributes to advancing machine vision techniques in addressing longstanding challenges in pavement crack detection. Future work is suggested for crack type classification following successful detection. Boateng K. T. A. et al. [8] proposes an astute discernible sound-based technique for motor driving road crack detection. Using an experimental setup, 13 highlights were extricated from crankshaft sound examples, with 7 selected through neighborhood component analysis (NCA). Machine learning models, including k-Nearest Neighbors (KNN) and Support Vector Machine (SVM), were employed for classification. The selected KNN and SVM models demonstrated high accuracy, with KNN outperforming at 98%. This innovative approach provides a promising solution for efficient and accurate engine crankshaft crack detection.

Busheska A. et al. [9] addresses the pressing need for sustainable building practices through the fusion of infrared thermography and machine learning. By employing Convolutional Neural Networks (DCNN), the study classifies three degrees of building breaks using images from different thermal cameras. Comparison of model accuracy across Multi-Spectral Dynamic (MSX), predictable, and current images, along with the analysis of fused images, contributes to a more efficient and accurate identification of building pathologies. This innovative approach supports sustainable construction practices by aiding stakeholders in assessing existing building conditions effectively. Yadav D. K. et al. [10] proposed cost-effective wireless slope monitoring system utilizing a sensor network for real-time tracking of slope movements in mining areas. Addressing the high costs of existing technologies like slope stability radar and LiDAR, the framework, planned on the Arduino environment with ZigBee for wireless communication, offers visible warnings of potential slope failures. Python is employed for information base and graphical UI implementation. The article emphasizes the economic efficiency and practicality of this solution for slope stability monitoring, vital in preventing hazards to lives and property in mining operations. Field data verification with a total station monitoring system adds credibility to the proposed system.

Wu H. et al. [11] examines the significance of Optical Fiber Sensors (OFS) in monitoring slope instabilities, crucial for ensuring safety in hilly, mining, and dam regions. OFS offers distinct advantages, like resistance to electromagnetic obstruction, distant transmission, and high sensitivity. The overview focuses on two OFS-based approaches: fiber Bragg grating based detecting and conveyed OFSs, including Brillouin optical time space reflectometry and Brillouin optical time area examination. The article highlights the technology's latest developments while tending to current difficulties in carrying out OFS for effective slant twisting observation. Qian Y. et al. [12] addresses the challenging control issues in a 5 DOF dual transport mounted crane arrangement, crucial for marine cargo transportation. Introducing an adaptive dynamic programming (ADP)-based optimal getting the hang of sliding mode regulator, the approach considers the system's nonlinearity, under actuation, and uncertainties from wave disturbances. Utilizing the Hamilton-Jacobi-Bellman equation and gradient attenuation algorithm, the proposed controller ensures stability. Concluding the examination of the

literature, this research describes its unique contributions to the domain, clearly differentiating it from prior methodologies.

- (i) In the proposed approach, crack detection using YOLO method is integrated with the machine learning approach as predictive model.
- (ii) The proposed model is compared with other competitive algorithms like YOLO V3 and YOLO V8.
- (iii) The performance of all the proposed approaches are compared by considering three performances metric, such as mean absolute error (MAE), root mean square error (RSME) and relative standard error (RSE).
- (iv) Two non-parametric statistical testing like Sign test, Wilcoxon Rank test and Friedman's test has been performed to investigate the dominance of proposed one over others.

The organization of the following sections is as follow. Section 3 describes the architecture of IoT. Methodology is explained in section 4, and section 5 illustrates the result analysis. Finally, in section 6 the conclusion and future scope of research is highlighted.

3. Research Design

3.1. Architecture Of IoT

The Internet of Things (IoT) has seamlessly integrated into our daily lives, connecting ordinary objects to the internet and facilitating the collection and sharing of data. To comprehend the functioning of this intricate network, it is essential to delve into the IoT architecture, a layered structure that orchestrates the fluid exchange of information between devices, networks, and applications. The IoT architecture can be broadly categorized into four interconnected layers, each playing a pivotal role in the system:

- Sensing Layer:

Serving as the foundational layer, it encompasses sensors and actuators embedded in diverse devices.

Sensors collect data, such as temperature, pressure, or movement, from the physical environment, while actuators respond to commands, controlling physical systems.

- Network Layer:

Responsible for communication, this layer employs various protocols like WiFi, Bluetooth, or cellular networks to facilitate data transmission between devices and gateways.

Gateways act as intermediaries, filtering and aggregating data before forwarding it to the subsequent stage.

- Data Processing Layer:

In this layer, collected data undergoes analysis and processing, either in cloud platforms or edge computing devices.

Tasks like filtering, aggregation, and analytics are performed to extract valuable insights from the raw data.

- Application Layer:

The final layer interfaces with users through applications built on web or mobile platforms.

These applications leverage the processed data to deliver specific services or functionalities, exemplified by smart home apps controlling lights or industrial dashboards monitoring manufacturing processes.

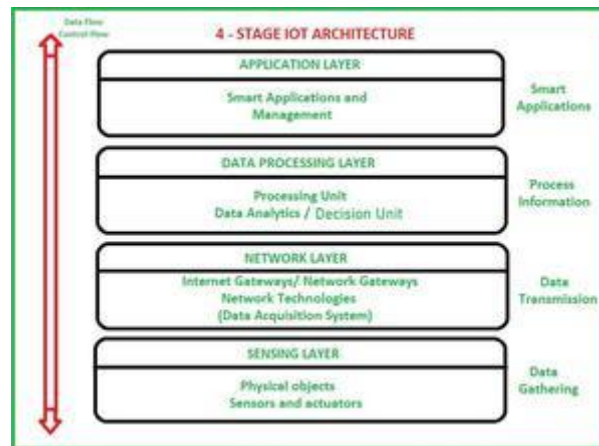


Fig.2. Stages of IoT Architecture

The seamless interaction between these layers facilitates a smooth flow of information, where sensors collect data, networks transport it, processing extracts insights, and applications deliver value [13].

Additional considerations:

- **Security:** Protecting data privacy and device integrity is crucial, requiring robust security measures throughout the architecture.
- **Scalability:** Systems need to adapt to accommodate a growing number of connected devices.
- **Interoperability:** Devices and protocols need to communicate seamlessly regardless of the manufacturer or network.

This knowledge empowers us to leverage its potential for improved efficiency, convenience, and innovation across various sectors. The system's scalability is vital to adapt to an increasing number of connected devices, and interoperability ensures seamless communication between devices and protocols regardless of manufacturer or network.

3.2. Architecture of YOLO Models

The YOLO (You Only Look Once) construction bears resemblances to GoogleNet. It incorporates twenty four convolutional layers, four max-pooling layers, and two fully connected layers, all input pictures are organized sequentially to analyze and derive pertinent structures (refer to fig 1). Nevertheless, it's crucial to acknowledge that the arrangement and structure of YOLO might differ based on the form and variation under consideration.

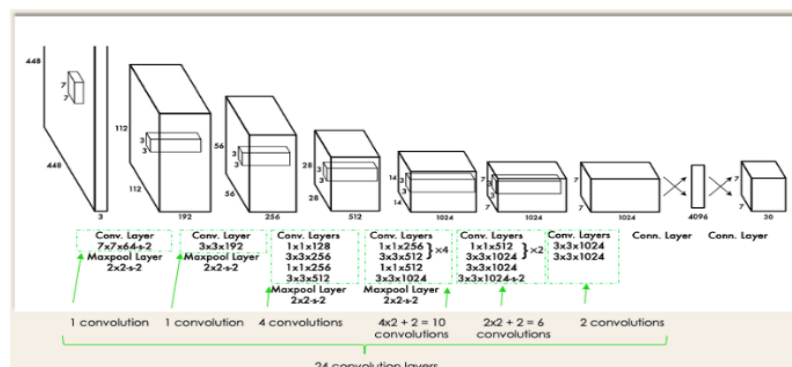


Fig. 3. YOLO Architecture

The YOLO model functions through the following steps:

- Commencing with a 1x1 convolutional layer, the architecture works to decrease the channel count in the input.

- Subsequently, a 3x3 convolutional layer is employed to produce a cuboidal result.
- The Rectified Linear Unit (ReLU) activation function is consistently useful throughout the architecture to present non-linearity, excluding the last layer, which utilizes a linear activation function.
- To optimize the model's output and counter over fitting, supplementary methods such as batch regularization and dropout are implemented. Batch regularization regulates the productivities of individual layer, enhancing working out steadiness and expediting convergence. Dropout arbitrarily omits a portion of neurons at the time of training, tumbling the model's dependence on explicit attributes and augmenting its simplification capability.

3.3. YOLO Models

It is crucial to remember that advances in deep learning and computer vision are still being made, and that since then, the YOLO model may have undergone revisions or releases. To make sure you are aware of the most recent developments about the YOLO model or any alternative deep learning models, it is always a good idea to review the most recent studies and available materials.

3.3.1. YOLO V3

YOLO V3 stands out as a widely acclaimed, rapid, and exact constant item discovery calculation that incorporates multiple-scale forecast and utilizes darknet53 as its main classifying algorithm. Darknet53, serving as the convolutional neural network for component determination, enables the discovery of multiple objects within a single frame and efficiently locates their positions in an image. In its operation, YOLO V3 divides the image into $n \times n$ grid cells, predicting multiple objects for each cell or set of cells. This involves considering cell probability, bounding box predictions across images, and selecting the most probable bounding box for a portion of picture. YOLO V3 accommodates various image outlines, such as 416 x 416, 608 x 608, etc., where the shape must be divisible by 32 without any remainder.

The input for YOLO V3 is accepted as a group of pictures with the form $(n \times w \times h \times c)$, where n signifies the number of pictures, w signifies the breadth of picture, h signifies the elevation of picture, c signifies the frequency of picture.

The algorithm employs a 1×1 kernel on the down-sampled picture at three dissimilar identification layers, generating an result shape of $(n \times n \times c)$, where $(n \times n)$ signifies the attribute map size subsequent to down-sampling, and c is calculated as $(b * (5 + c))$, with b representing the quantity of recognition layers and c representing the number of classes. In the proposed system, this results in $(3 * (5 + 13))$, reflecting the attempt to forecast 13 classes across three dissimilar layers.

3.3.2. YOLO V5

YOLO V5 stands out as the latest and most advanced entity recognition procedure, employing a improved Convolutional Neural Network (CNN) model with the Cross Stage Partial CSP-Darknet as the main, complemented by a Spatial Pyramid Pooling (SPP) layer. The design is structured into three key components. In the initial phase, CSP-Darknet is employed to extract essential features from input images. The subsequent segment, referred to as Neck and YOLO V5, utilizes PANet, a pyramid method facilitating the recognition of objects across various scales. The third segment, known as the Head, is accountable for the final identification of substances within the input, marking crucial boxes on pictures to formulate the conclusive result, complete with class stamp sand names.

Building upon the achievements of its predecessors, YOLO V5 introduces various enhancements and new features. Diverging from YOLO, YOLO V5 adopts intricate design termed Efficient, based on the Efficient Net system design (depicted in fig. 4). This added multifaceted design enables YOLO V5 to attain heightened accurateness and improved overview across a broader spectrum of object groups.

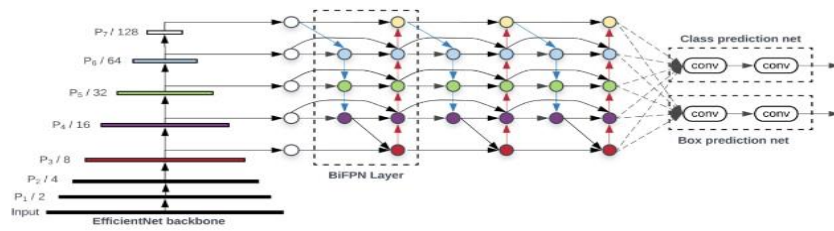


Fig. 4. Architecture of the EfficientDet model

3.3.3. YOLO V8

The current addition to the YOLO family is YOLO V8. YOLO, which stands for "You Only Look Once," is a series of models known for their capacity to accurately predict each item in a picture through a solitary forward pass. These models are being pre-trained on extensive datasets like COCO and Image Net; possess a unique combination of being both Master and Student. They excel in providing precise predictions for classes they are pre-trained on (master ability) while demonstrating adaptability to learn new classes with relative ease (student ability).

YOLO models offer advantages such as faster training, the capacity to achieve better precision with minor model sizes, and compatibility with single GPU training, making them available for designers. YOLO V8, the modern recapitulation as of early 2023, introduces significant changes from its predecessors, including anchor-free recognition, the incorporation of C3 convolutions, and the implementation of montage expansion.

3.4. Machine learning Models used for accuracy prediction

These machine learning models encompass a spectrum of approaches, ranging from a fundamental baseline (ZeroR) to more intricate methodologies like RBFR and Kstar, which are adept at handling non-linear relationships and instance-based learning. Random Forest, characterized by its ensemble learning approach, enhances accuracy and generalization across diverse datasets, making it a versatile choice.

3.4.1. ZeroR:

- Overview: ZeroR serves as a basic and uncomplicated machine learning model often employed as a benchmark for comparison against more intricate models.
- Functionality: Its operational principle involves forecasting the utmost predominant class in the training for all cases in the test dataset, effectively disregarding any features. This simplistic approach is valuable for establishing a foundational performance metric against which more advanced models can be evaluated.

3.4.2. RBFR (Radial Basis Function Regression):

- Overview: RBFR stands as a form of regression model utilizing radial basis functions as basic functions for interpolation.
- Functionality: Particularly effective in handling non-linear data, RBFR seeks to approximate a target function by combining radial basis functions with coefficients established during the training phase. This model is commonly employed in scenarios where the relationship between input features and the target variable is intricate and not easily captured by linear models.

3.4.3. Kstar:

- Overview: Kstar functions as a versatile machine learning procedure appropriate to both classification and regression.
- Functionality: Classified as a lazy learner, Kstar defers the learning process until predictions are required. It employs an instance-based approach, where predictions rely on the similarity of instances in the training set to the new instance. Kstar is distinctive for its utilization of an entropy-based approach for attribute selection, making it particularly advantageous in situations involving both discrete and continuous attributes.

3.4.4. Random Forest (RF):

- Overview: Random Forest represents an collective learning technique constructing several decision trees during the training phase.

- **Functionality:** Individual tree within the forest autonomously estimate forecasts, and the result and decision is evaluated through a majority voting or average calculation, depending on whether it is a classification or regression task. Renowned for its robustness, accuracy, and ability to handle extensive datasets with numerous features, Random Forest mitigates over fitting by aggregating predictions from multiple trees.

4. Methodology

4.1. Flow of work

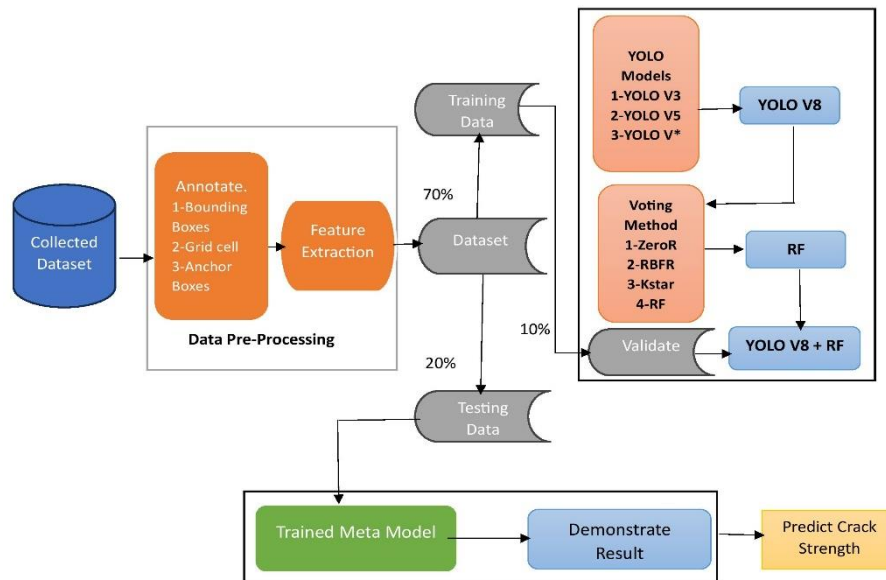


Fig.5. flow work of proposed model

4.2. Data Collection

We collect the dataset for our research both manually and IoT devices. Some devices which we are used for this are discuss below. Structural deformity detection in mining operations is a decisive factor of ensuring the security and stability of mining infrastructure, such as tunnels, shafts, and support systems. Traditional methods of detecting structural deformities involve a range of techniques and technologies aimed at identifying changes in the shape, alignment, and integrity of mining structures [14].

Here are some traditional methods used for structural deformity detection in mining operations:

- Visual Inspection:** Regular visual inspections by trained personnel remain one of the simplest and most common methods for detecting visible deformities in mining infrastructure. This involves walking through tunnels, shafts, and other structures to identify any visible cracks, shifts, or signs of instability.
- Manual Measurements:** Traditional measurement techniques involve using basic tools like measuring tapes, levels, and plumb bobs to quantify deformities. Measurements of distances, angles, and shifts are compared to baseline values to identify changes.
- Surveying:** Surveying instruments such as overall positions and transits can be used to quantify positions and remoteness with better accuracy. These measurements are compared to previous surveys to detect changes in the shape and alignment of structures.
- Inclinometers:** Inclinometers are devices that measure the angle of inclination or tilt of a structure. They are commonly used to monitor the stability of slopes, tunnels, and other underground structures.
- Extensometers:** Extensometers are devices used to measure changes in the length or displacement of structures. They are often used to monitor the movement of tunnel walls, roof, and floors.
- Strain Gauges:** Strain gauges are sensors that measure the strain or deformation of materials under load. They can be attached to structural elements to monitor stress changes and identify potential deformities.

- vii. Ground Penetrating Radar (GPR): GPR is a geophysical technique that uses detector beats to represent the subarea. It will help to distinguish voids, cracks, and other deformities in the rock surrounding mining structures.
- viii. Acoustic Emission Monitoring: This method involves detecting and analyzing the acoustic emissions produced by the cracking or deformation of materials. It can help identify areas of stress concentration and potential structural issues.
- ix. Micro seismic Monitoring: Micro seismic monitoring involves detecting and analyzing micro seismic events (tiny seismic waves) caused by rock deformation. It can provide insights into rock stability and potential hazards.
- x. Photogrammetry: Photogrammetry uses photographs of structures taken from multiple angles to create 3D models. Changes in the models over time can indicate deformities.
- xi. Geodetic Monitoring: Geodetic techniques involve measuring changes in the Earth's surface using satellite-based systems like GPS. They can provide accurate information about ground movement and deformation.

In below figure 6 given the sample dataset which we are collected

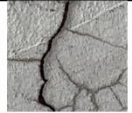
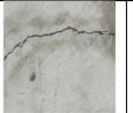
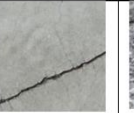
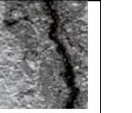
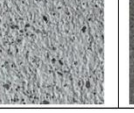
Type of Crack	Sample Images					
Positive						
Negative						

Fig.6. Sample of collected dataset

4.3. Data Pre-processing and Annotation

The goal of the YOLO algorithm is to anticipate both an object's class and the hopping box which indicates where the object is located in the input picture. It uses four digits to identify each bounding box: The bounding box's center (bx, by), The box's width (bw), The box's height (bh) [15].

Furthermore, YOLO forecasts the probability of the prediction P_c in addition to the consistent c numeral for the anticipated class.

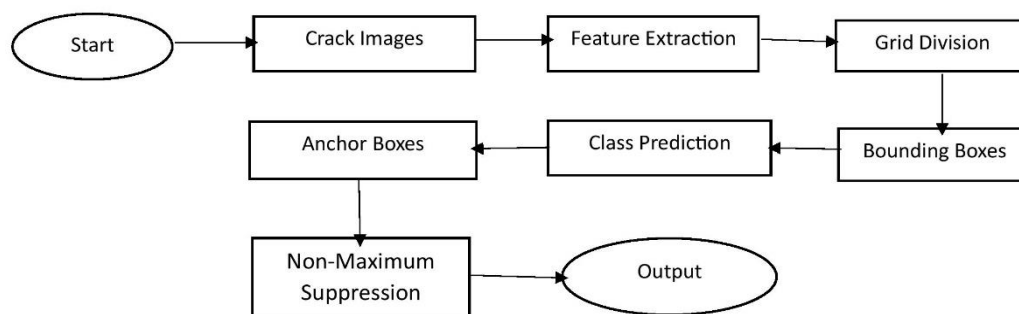


Fig.7. Data preparation

4.3.1 Feature Extraction: The image passes through a convolutional neural network (CNN) to extract features. YOLO uses various layers of convolution, pooling, and activation functions to capture hierarchical features from the image.

4.3.2 Grid Division: The image is divided into a grid of cells. Each cell is responsible for detecting objects within its boundaries.

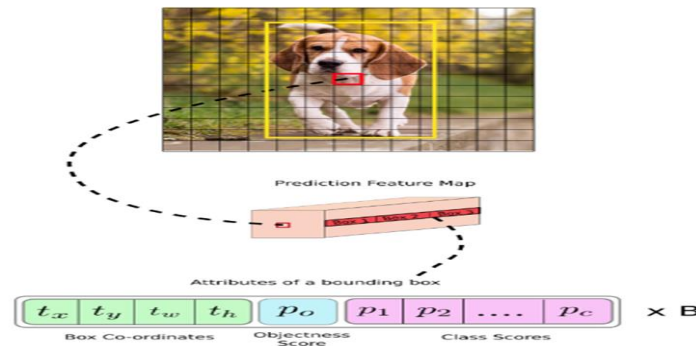


Fig.8. Grid Cell over image data

It makes 13 X 13 number of grid cell over the interested image data set.

4.3.3 Bounding Box Prediction: For each grid cell, YOLO predicts bounding boxes that enclose the objects present in that cell. A standard classification task would involve an image running through a Convnet with multiple layers in which vector features are fed into a Soft Max unit for example that outputs the predicted class (Object categories that the algorithm is trying to detect i.e cars, trees, pedestrians). Object detection models such as YOLO works by splitting images into a grid of cells where each grid cell is responsible for predicting a bounding box if the center of the bounding box lies in the cell. It will then output the predicted class, coordinates of the bounding box as shown below. These bounding boxes are defined by their center coordinates, width, height, and confidence score.



Fig.9. bounding boxes on interested image

4.3.4 Class Prediction: YOLO also predicts the probability distribution over different classes for each object present in the cell.

4.3.5 Anchor Boxes (YOLO v2 and later): YOLO v2 and subsequent versions introduce the concept of anchor boxes. These are pre-defined shapes that help the model predict bounding box dimensions more accurately. Here's where the idea of anchor boxes can be used. Looking at the image below, notice how both the person and car midpoints lie within the grid cell. (For simplicity's sake I split the image into a 3 by 3 grid)

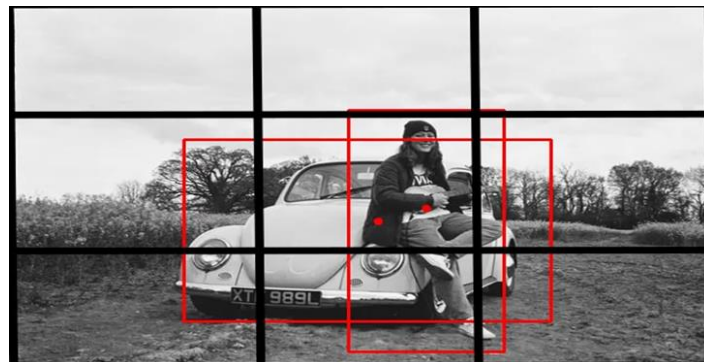


Fig.10. Anchor box apply over image

4.3.6 Non-Maximum Suppression: After predictions are made for all cells, a post-processing step called Non-Maximum Suppression (NMS) is applied. NMS eliminates duplicate and low-confidence predictions, retaining only the most confident bounding boxes for each object.

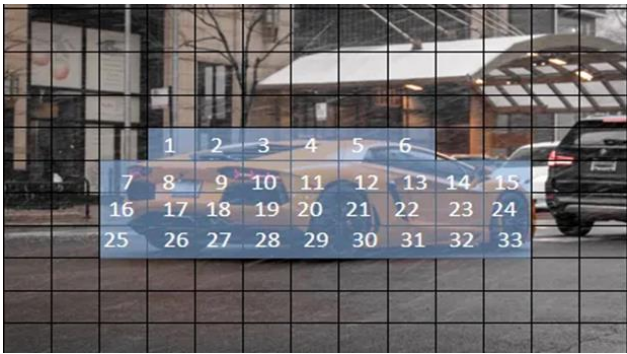


Fig.11. NMS over interested chosen area

Output: The final output of the YOLO model is a set of bounding boxes along with their corresponding class labels and confidence scores.

The key advantages of the YOLO architecture include its speed and efficiency. Since the model processes the entire image in one pass, it is inherently faster than other approaches that perform multiple passes for object detection. This makes YOLO suitable for applications where real-time processing is crucial.

4.3.7 Evaluation Metrics

For the comparison of different models few evaluation metrics are used such as MAP (mean Average Precision). Performance measurement metrics such as precision and recall are used for the object recognition techniques [17]. In below figure 12 and 13, we graphically presented the metrics which we are found at time of training.

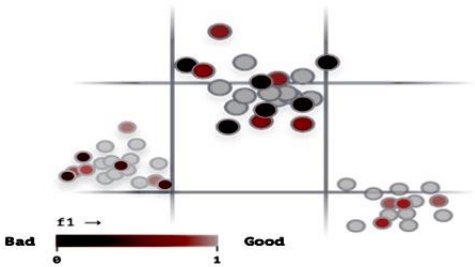


Fig.12. vector analysis

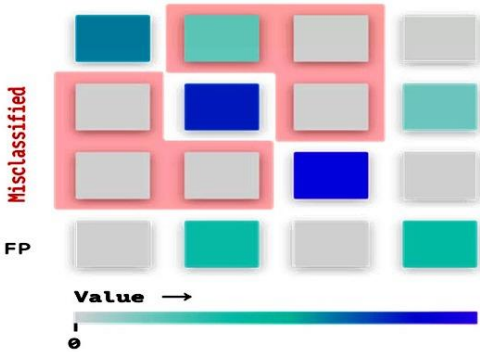


Fig.13. confusion matrix

5. Simulation results and discussion

Three varieties of the YOLO (You Only Look Once) techniques are used, namely YOLO V3, YOLO V5, and YOLO V8, to accomplish the recognition task. Above-mentioned models have been nominated for their substantial improvements in crack recognition.

Table I comparison table of YOLO Models

Models	mAP	Precision	Recall
YOLO v3	82.6	72.2	75.9
YOLO v5	88.41	87.9	63.5
YOLO v8	95.7	93.3	86.5

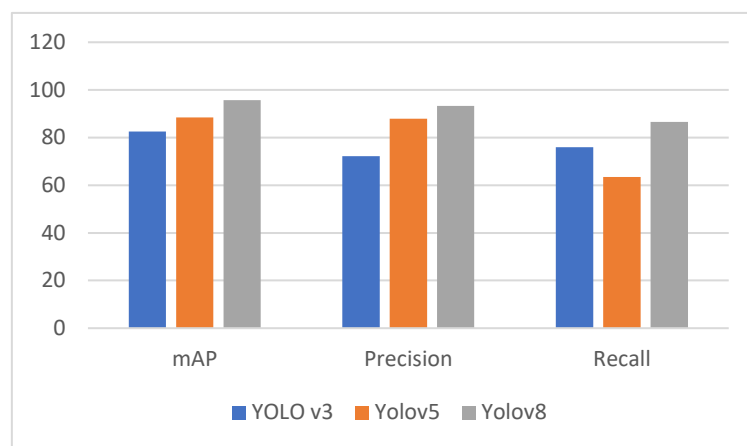


Fig. 14. Graphical presentation of comparison of YOLO models

After successfully found that the proposed model gives the best recognition precision of cracks in bridges. For check the strength of that construction we used collected numeric data which are collected by IoT devices are trained by ML models i.e., ZeroR, RBFR, Kstar and RF for prediction accuracy over Pressure, Thrust, displacement and response time. We found Random Forest gives the best accuracy and low error values. ZeroR gives 89.28%, RBFR gives 88.44% Kstar gives 86.99%, RF gives 94.23%.

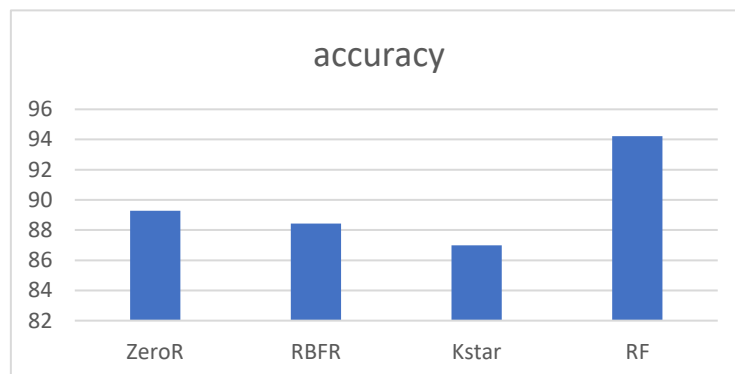
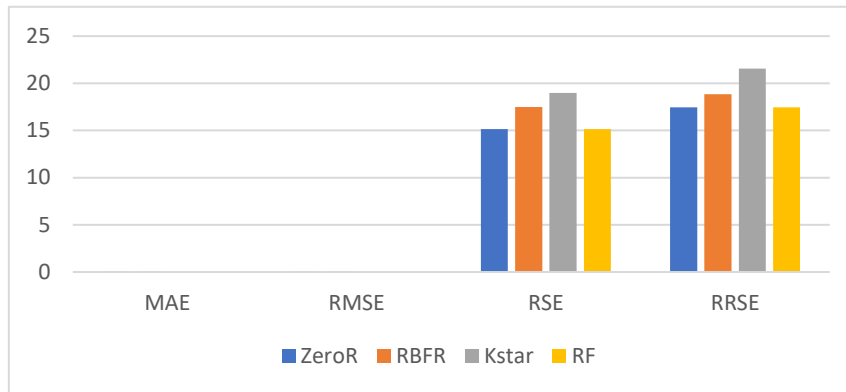


Fig.15. Accuracy of taken ML models

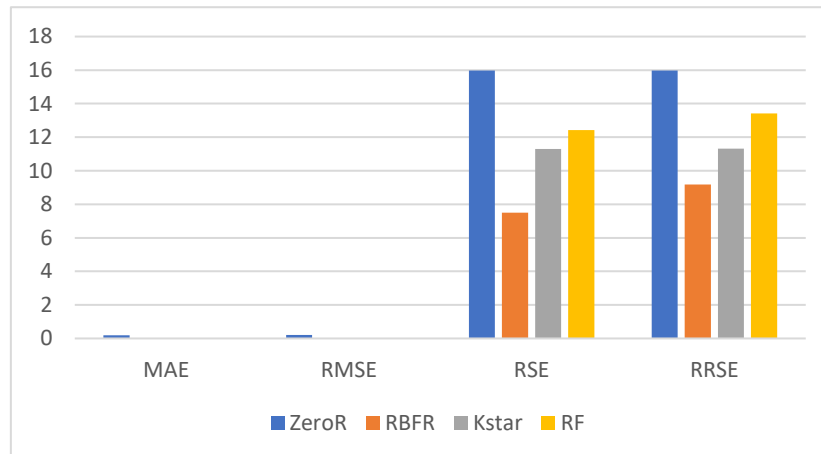
The accuracy and error values are stored in below table II, III and IV and graphically presentation in figure 16 and 17 and 18.

Table II Error values of ML models over pressure datasets.

Models	MAE	RMSE	RSE	RRSE
ZeroR	0.015	0.023	15.14	17.45
RBFR	0.017	0.025	17.49	18.85
Kstar	0.019	0.028	18.99	21.55
RF	0.015	0.023	15.14	17.45

**Fig. 16 graphical representation error values of taken models over Pressure datasets****Table III Error values of ML models over Thrust and displacement datasets**

Models	MAE	RMSE	RSE	RRSE
ZeroR	0.183	0.208	15.98	15.98
RBFR	0.019	0.014	7.5	9.17
Kstar	0.02	0.023	11.29	11.31
RF	0.02	0.022	12.42	13.42

**Fig. 17 graphical representation error values of taken models over Thrust and displacement datasets****Table IV Error values of ML models over responds time datasets**

Models	MAE	RMSE	RSE	RRSE
ZeroR	0.33	0.76	65.74	65.7
RBFR	0.18	0.37	55.72	48.86
Kstar	0.07	0.39	23.92	51.98
RF	0.099	0.11	2.98	14.46

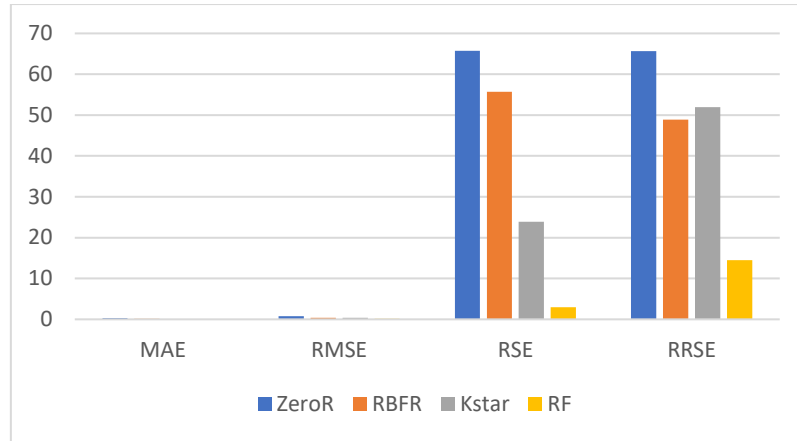


Fig. 18 graphical representation error values of taken models over respond time datasets

All the three YOLO techniques have considered various randomly initialized parameters, and hence, the output obtained from simulation is not consistent for every run. In our research work, for conducting a fair comparison of performance, we have performed Sign test and the Wilcoxon Signed rank test which are two standard statistical tests on different approaches pairwise. The Sign test and the Wilcoxon Signed rank test are implemented to determine the dominance of the proposed YOLOv8 algorithm. We have run all the three YOLO algorithm 25 times each, and the performance metrics have been noted for each run. For accomplishing significant levels of $\alpha=0.05$ and $\alpha=0.01$ different approaches are executed for 5 to 25 times. The minimum number of wins mandatory for achieving these significance level is shown in Table V.

Table V. For $\alpha=0.05$ and $\alpha=0.01$ level of significance, minimum number of wins required

No. of cases	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
$\alpha = 0.05$	5	6	7	7	8	9	9	10	10	11	12	12	13	13	14	15	15	16	17	18	18
$\alpha = 0.01$	5	6	6	7	7	8	9	9	10	10	11	12	12	13	13	14	14	15	16	16	17

Table VI illustrates the comparison among the algorithms by considering the MAE as a winning parameter after applying proposed and other approaches. It is observed that the YOLOv8 algorithm is dominating the other two YOLOv5 and YOLOv3 with a significance level of $\alpha=0.05$, as shown in Table VI.

Table.VI. Using MAE as the winning parameter, the comparison among proposed and others two competitive algorithm.

YOLOv8	YOLOv5	YOLOv3
Wins (+)	19	17
Loss (-)	1	3
Detected difference	$\alpha = 0.05$	$\alpha = 0.05$

By considering the same RSE and PSNR value, the p-value and h-value as the triumphant parameters are depicted in Table VII and VIII.

Table VI. Wilcoxon Sign test using RSE as winning parameters

Comparison	p-value	h-value
YOLOv8 to YOLOv5	0.0078	1
YOLOv8 to YOLOv3	0.0058	1

Table VII. Wilcoxon signed test using PSNR value as winning parameter

Comparison	p-value	h-value
YOLOv8to YOLOv5	0.0018	1
YOLOv8 to YOLOv3	0.0027	1

Additionally, the Friedman test which detects the dominance between the two algorithms is depicted in Tables VIII and IX.

Table VIII. Ranking table for the Friedman test

Methods	YOLOv8	YOLOv5	YOLOv3
Mean Ranks	15.9	5.3	16.6

Table IX. Parameter for the Friedman test

Source	Sum of Squares	Degree of Freedom	Mean Square	Chi-Square	Critical value(p)
Column	141.8	3	44.261	54.61	1.652E-12
Error	61.8	69	0.6812		
Total	187	102			

Friedman's statistical test has also been employed to obtain the dominance level of the YOLOv8 approach over other. This is one of the approach which is relied on Annova-2, and suitable to investigate both the significance and the repeatability of the results. This test is performed on the MAE value which are collected during the different experiments. The values of Friedman's test are depicted in Tables X and XI respectively. It is observed that the usefulness of the proposed YOLOv8 based approach is outplays others. The null hypothesis is utilized here for obtaining the results.

Table X. Average ranking of denoising algorithms on the basis of Friedman's test

Algorithms	WIENER	YOLOv3	YOLOv5	YOLOv8
Ranking	11	12.04	11	6

Table XI. Friedman Statistical test parameters using MAE

Source	Sum of square (SS)	Degree of freedom (df)	Mean square (MS)	Chi-square	Critical value (p)
Columns	569	6	121.67	12.25	0.0024
Error	599	27	32	-	-
Total	1276	39	-	-	-

The proposed approach takes a reasonable amount of time that required to train the parameters that is necessary for any crack detection approach. It is a two-stage algorithm for obtaining optimum solution, and is a parameter specific approach. The proposed approach depends on initial parameter for finding solution. Apart from the efficacy in simulation output of the proposed approach, it is simple in implementation as well as efficient in time complexity. From simulation studies it observes that it takes 0.0612 second and 3.9 second to complete iteration and one epoch respectively.

6. Conclusion and future works

The recommended model, which can identify connection crashes more accurately and swiftly, combines DCGAN and YOLOv8. The training environment and related theories have been established, and the DCGAN and YOLOv8 neural networks' respective parameters and organizational structures have been established. In this study, the model's performance is investigated and contrasted with the larger dataset's results. The trained DCGAN can quickly make countless counterfeit bridge crack images by learning the features of the cracks, which is used to increase the size of the real dataset. The outcome shows that when YOLOv8 is generated on the expanded dataset, it demonstrated the same accuracy as when it is prepared on the initial dataset, which provides an entirely novel perspective for support cost control and observation of the large concrete structure. The simulation results depict that the proposed approach gives superior result as compared to others. Moreover, the statistical analysis shows the dominance of proposed model over others.

It is observed that, it requires high-level data preparation methods, and any other advanced optimization approach for updating the weights of neural. As there is huge demand for SHM, it grows exponentially as per market trends. It is projected to grow at a 14.5% CAGR between 2022 and 2032. This growth is expected to be driven by application in building and construction, where construction assumes a significant part in guaranteeing safety and minimizing maintenance costs. In future we can add sensor technology, data analytics for real-time monitoring and predictive maintenance.

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Conflicts of interest

The authors declare no conflict of interest.

Authors contribution statement

The corresponding author has performed design of proposed algorithm, comparative analysis. Dr Pani has performed related work and future scope.

Data availability

The data is not available publicly. It can be produced on demand by the corresponding author.

REFERENCES

- [1] Cao Vu Dung, Le Duc Anh, "Autonomous concrete crack detection using deep fully convolutional neural network", *Automation in Construction*, Volume 99, 2019, Pages 52-58, ISSN 0926-5805, <https://doi.org/10.1016/j.autcon.2018.11.028>.
- [2] Fen Fang, Liyuan Li, Ying Gu, Hongyuan Zhu, Joo-Hwee Lim, A novel hybrid approach for crack detection, *Pattern Recognition*, Volume 107, 2020, 107474, ISSN 0031-3203, <https://doi.org/10.1016/j.patcog.2020.107474>.
- [3] Zhang L, Shen J, Zhu B. A research on an improved Unet-based concrete crack detection algorithm. *Structural Health Monitoring*. 2021;20(4):1864-1879. doi:10.1177/1475921720940068
- [4] Xu, H.; Su, X.; Wang, Y.; Cai, H.; Cui, K.; Chen, X. Automatic Bridge Crack Detection Using a Convolutional Neural Network. *Appl. Sci.* 2019, 9, 2867. <https://doi.org/10.3390/app9142867>
- [5] Raza Ali, Joon Huang Chuah, Mohamad Sofian Abu Talip, Norrima Mokhtar, Muhammad Ali Shoaib, Structural crack detection using deep convolutional neural networks, *Automation in Construction*, Volume 133, 2022, 103989, ISSN 0926-5805, <https://doi.org/10.1016/j.autcon.2021.103989>.
- [6] T. H. Dinh et al., "Toward Vision-Based Concrete Crack Detection: Automatic Simulation of Real-World Cracks," in *IEEE Transactions on Instrumentation and Measurement*, vol. 72, pp. 1-15, 2023, Art no. 5032015, doi: 10.1109/TIM.2023.3328076.
- [7] A. R. Rababaah, "A Deep Learning based Process Model for Crack Detection in Pavement Structures," 2022 9th International Conference on Computing for Sustainable Global Development (INDIACom), New Delhi, India, 2022, pp. 1-6, doi: 10.23919/INDIACom54597.2022.9763286.
- [8] K. T. Asamoah Boateng, H. Nunoo-Mensah, J. Ohene-Akoto, P. E. Adjei and K. O. Boateng, "Detection of Cracks in Crankshaft Using an Intelligent Audible Sound-Based Non-Destructive Method," 2021 Asian Conference on Innovation in Technology (ASIANCON), PUNE, India, 2021, pp. 1-6, doi: 10.1109/ASIANCON51346.2021.9545038.
- [9] A. Busheska, N. Sabella, N. Almeida and E. Rocha, "Machine Learning and Thermography Applied to the Detection and Classification of Cracks in Buildings," 2023 IEEE Conference on Technologies for Sustainability (SusTech), Portland, OR, USA, 2023, pp. 247-251, doi: 10.1109/SusTech57309.2023.10129614.
- [10] D. K. Yadav, G. Karthik, S. Jayanthu and S. K. Das, "Design of Real-Time Slope Monitoring System Using Time-Domain Reflectometry With Wireless Sensor Network," in *IEEE Sensors Letters*, vol. 3, no. 2, pp. 1-4, Feb. 2019, Art no. 2500304, doi: 10.1109/LSSENS.2019.2892435.
- [11] H. Wu, Y. Guo, L. Xiong, W. Liu, G. Li and X. Zhou, "Optical Fiber-Based Sensing, Measuring, and Implementation Methods for Slope Deformation Monitoring: A Review," in *IEEE Sensors Journal*, vol. 19, no. 8, pp. 2786-2800, 15 April 2019, doi: 10.1109/JSEN.2019.2891734
- [12] Y. Qian, D. Hu, Y. Chen and Y. Fang, "Programming-Based Optimal Learning Sliding Mode Control for Cooperative Dual Ship-Mounted Cranes Against Unmatched External Disturbances," in *IEEE Transactions on Automation Science and Engineering*, vol. 20, no. 2, pp. 969-980, April 2023, doi: 10.1109/TASE.2022.3182720.
- [13] Kong, Xiangxiong, and Ronny Garrett Hucks. "Preserving our heritage: A photogrammetry- based digital twin framework for monitoring deteriorations of historic structures." *Automation in Construction* 152 (2023): 104928.
- [14] Wang, Dapeng, and Wenda Zhang. "Damage detection combining principal component analysis and deep convolutional neural network with dynamic response from FBG arrays." *Journal of Civil Structural Health Monitoring* 13.1 (2023): 101-115.
- [15] Yessoufou, Fadel, and Jinsong Zhu. "One-class convolutional neural network (OC-CNN) model for rapid bridge damage detection using bridge response data." *KSCE Journal of Civil Engineering* 27.4 (2023): 1640-1660.
- [16] Qureshi, Kashif Naseer, et al. "Neurocomputing for internet of things: object recognition and detection strategy." *Neurocomputing* 485 (2022): 263-273.
- [17] Pan, Xi, and Tingsheng Zhao. "Pressure deviation monitoring and early warning in large integrated tower crane support system for super high-rise buildings." *Measurement: Sensors* 31 (2024): 101019.